CLIENT UPDATE

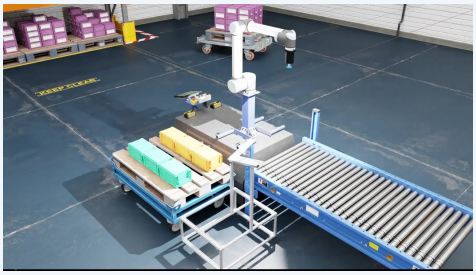
Isaac robotics workshop

Two demos • 60 minutes • Beginner-friendly

Both demos have been completed and verified in Isaac simulation. Demo recordings, illustrated workbooks and repeatable workshop files are ready. The session introduces robotics through clear images and step-by-step mouse and keyboard actions.

Demo 1 | Physical sorting

05:00–25:00 • UR10 with suction gripper



Set up the conveyor, robot and six containers. Explain robot joints, inverse kinematics (IK) and the orange-to-A / green-to-B sorting rule. Follow pickup, transfer and release; check that all containers settle.

Verified result: 6/6 sorted correctly — three in each group.

Demo 2 | RL gear insertion

25:00–56:00 • Eight parallel Frankas



Close-up: gear insertion

Build the yellow/white gear workbench with NIST reference assets. Randomize eight robot starts and generate camera data. Review real reinforcement learning (RL) using PPO, starting from a proven policy. Test and reload the saved policy on one simulated Franka.

Recorded results: 12 PPO updates / 3,072 new transitions; 8/8 fresh evaluation trials and 2/2 single-Franka deployment trials.

Materials prepared for repeat use

Step-by-step videos, large-print workbooks, runnable workshop files, trained and deployment policies, and 16 synthetic camera captures with RGB images, depth and object labels.

Both demonstrations and policy deployment run in simulation.

SESSION PLAN

Exact 60-minute agenda

Times below are minutes from the start of the session.

From → To	Topic and what we will cover
00:00–05:00	**Welcome & robot basics.** Links, joints, grippers and IK: how the arm reaches a target.
05:00–10:00	**Demo 1 — asset setup.** Open Isaac; add the conveyor, UR10, suction gripper and receiving pallet.
10:00–15:00	**Demo 1 — sorting rule.** Orange to A; green to B. Explain detect, pick, move and release.
15:00–25:00	**Demo 1 — physical sorting.** Walk through the complete six-box recording and verify all placements.
25:00–31:00	**Demo 2 — scene & randomization.** Inspect the gear workbench and eight independently randomized robot starts.
31:00–36:00	**Demo 2 — synthetic data & Gym.** Generate RGB, depth and labels; explain observations, actions and rewards.
36:00–47:00	**Demo 2 — RL learning.** Review recorded PPO fine-tuning, the shared policy, rewards and saved weights.
47:00–51:00	**Demo 2 — evaluation.** Inspect eight fresh trials and check stable insertion results.
51:00–56:00	**Demo 2 — policy deployment.** Export and reload the policy on one simulated Franka; watch it insert.
56:00–60:00	**Recap & questions.** Review the workshop files, repeat steps and discuss next steps.

How the hour stays on schedule: Live walkthroughs are combined with selected footage from the verified sorting, training and test runs. Isaac, assets and checkpoints are prepared before the session; full training can be repeated afterwards using the supplied commands.

Task scope: Insertion starts with the gear held. The NIST board is a reference asset beside the Factory insertion fixture. Camera data is a separate vision exercise; PPO learns from simulation state.